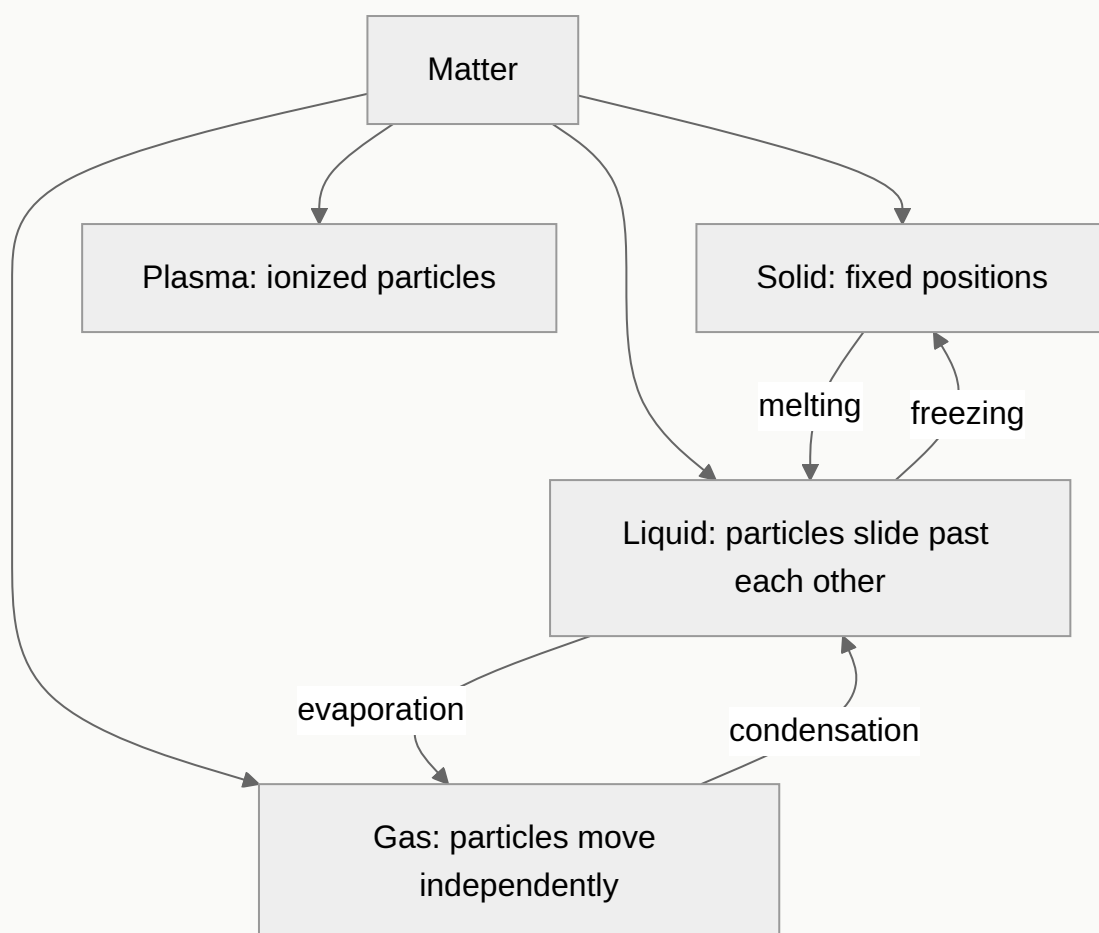


Matter

Matter is anything that has mass and occupies space. Every object you can touch, weigh, or contain in a jar — air, water, rock, your own body — qualifies as matter. According to atomic theory, all matter is composed of atoms: discrete particles of roughly 10^{-10} m in diameter that combine to form molecules and, ultimately, every observable substance. Matter exists in distinct physical states — solid, liquid, gas, and plasma — depending on how tightly its particles are bound and how much energy they carry. In a solid, atoms vibrate in fixed positions; in a liquid, they move past one another while staying in contact; in a gas, they move independently and spread to fill available space.

A useful way to apply this concept is through the idea of conservation of mass: in any physical or chemical process, matter is neither created nor destroyed, only rearranged. Consider dissolving 5 g of salt in 100 g of water. The resulting saltwater solution has a mass of exactly 105 g — the salt hasn't vanished, it has dispersed at the atomic level among the water molecules. This principle lets chemists predict the outcomes of reactions: if 12 g of carbon reacts completely with oxygen to form carbon dioxide, the mass of CO_2 produced must equal the combined mass of carbon and oxygen consumed, since no atoms are lost.

This idea becomes a practical problem-solving tool in chemistry. Suppose 10 g of ice melts into liquid water and then evaporates into vapor. Although the substance changes state — appearance, volume, and density all shift dramatically — the mass remains 10 g throughout, because no atoms have been added or removed, only their arrangement and spacing have changed. Recognizing this distinction between a change in state (physical change, same substance) and a change in composition (chemical change, new substances formed) is essential for correctly tracking matter through any process, from a melting ice cube to a rusting nail.



The four common states of matter and the transitions between them, driven by changes in particle energy and arrangement.